

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 30%

QUESTIONS: 5

Total Marks:25

Annexure A

Comparative Analysis

Code of Conduct:

*I M.HARINI (192424011) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: M. harini

Annexure A

Comparative Analysis

1. Star topology is generally more expensive than bus topology because it requires a central device such as a switch or hub and more cabling to connect each node individually. In contrast, bus topology uses a single backbone cable, making it low-cost and easier to install. However, in terms of reliability, star topology is superior because failure of one node does not affect others, whereas a failure in the main bus cable can bring down the entire network. Performance is also better in star topology since data transmission is managed through a central device, reducing collisions, while bus topology suffers from performance degradation as more devices are added due to increased data traffic on the shared medium.
2. Mesh topology offers very high fault tolerance because every node is connected to multiple other nodes, providing multiple paths for data transmission. If one link fails, data can still be routed through alternate paths without disrupting communication. In contrast, star topology has moderate fault tolerance; while failure of individual nodes does not affect the network, failure of the central hub or switch can cause the entire network to stop functioning. Therefore, mesh topology is more robust and reliable in critical environments compared to star topology.
3. Bus topology is simple and easy to install because it uses a single backbone cable with minimal connections, making it suitable for small networks. On the other hand, mesh topology is highly complex to install as each node must be connected to every other node, requiring a large number of cables and ports. This complexity increases significantly as the number of devices grows, making mesh topology time-consuming and costly to implement compared to the straightforward setup of bus topology.
4. Hybrid topology is highly scalable because it combines two or more different topologies (such as star, mesh, or bus), allowing the network to expand flexibly according to organizational needs. New segments can be added without affecting the existing network structure. In comparison, star topology is also scalable but to a limited extent, as it depends on the capacity of the central device (switch or hub). Once the central device reaches its port limit, expansion requires additional hardware. Thus, hybrid topology provides greater scalability than star topology.
5. In terms of maintenance, star topology is relatively easy to manage because issues can be quickly identified and isolated to a specific node or cable. Mesh topology, while highly reliable, is difficult to maintain due to its complex structure and large number of connections, making troubleshooting time-consuming. Bus topology is harder to maintain than star because faults in the backbone cable can disrupt the entire network and are difficult to locate. Hybrid topology has moderate maintenance complexity; while it benefits from flexibility, managing multiple combined topologies requires careful monitoring and skilled administration.

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Concepts	15%	Demonstrates complete and in-depth understanding of all concepts being compared	Good understanding with minor gaps	Basic understanding; some concepts unclear	Poor understanding of concepts
Comparison Depth	20%	Provides detailed, insightful, and meaningful comparisons across multiple aspects	Adequate comparison with relevant points	Limited comparison; lacks depth	Minimal or incorrect comparison
Criteria Selection	10%	Selects highly relevant and appropriate comparison parameters	Mostly relevant criteria	Some irrelevant or missing criteria	Poor or no criteria selection
Analytical Skills	15%	Strong analysis with logical reasoning and clear justification	Good analysis with some justification	Basic analysis with weak reasoning	No clear analysis
Organization & Structure	10%	Well-organized, clear, and logically structured	Generally organized	Somewhat disorganized	Poorly structured
Use of Examples / Evidence	10%	Uses strong, relevant examples or data to support comparison	Uses some examples	Limited examples	No supporting examples
Clarity of Presentation	10%	Very clear, concise, and easy to understand	Mostly clear	Somewhat unclear	Difficult to understand
Conclusion & Insights	10%	Insightful conclusion with strong justification	Clear conclusion	Basic conclusion	No or weak conclusion

Comparative Analysis of Network Topologies

Introduction

A computer network is a collection of interconnected computers, communication devices, and other networking equipment that communicate with each other to share data, resources, applications, and services. The arrangement or structure in which these devices are connected is known as network topology. Network topology describes both the physical arrangement of devices and the logical manner in which data moves between them.

Network topology plays an important role in determining the performance, reliability, scalability, cost, security, installation complexity, and maintenance requirements of a network. Selecting an appropriate topology is essential because different organizations have different requirements. A small office may require a simple and inexpensive network, while a large enterprise or critical infrastructure may require high reliability and multiple backup communication paths.

The major types of network topologies include Bus, Star, Ring, Mesh, Tree, and Hybrid topology. Among these, Star, Bus, Mesh, and Hybrid topologies are particularly important because they represent different approaches to network design.

This comparative analysis focuses on five major aspects:

1. Star topology compared with Bus topology
2. Mesh topology compared with Star topology
3. Bus topology compared with Mesh topology
4. Hybrid topology compared with Star topology
5. Maintenance comparison among different topologies

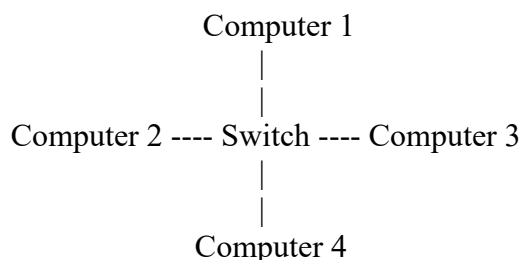
The analysis also considers cost, reliability, fault tolerance, performance, scalability, installation, troubleshooting, and practical applications.

1. Star Topology vs. Bus Topology

1.1 Star Topology

In a Star topology, all computers and network devices are connected to a central networking device. The central device is usually a network switch, although hubs were commonly used in older networks. Each node has a separate connection to the central device. When one computer wants to communicate with another computer, the data normally passes through the central switch.

Basic structure

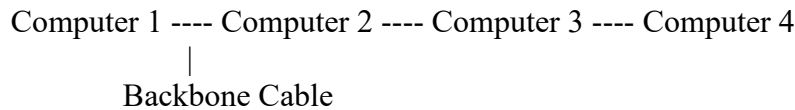


The central switch acts as the main point of communication. Modern Ethernet LANs commonly use a physical star arrangement.

1.2 Bus Topology

In a Bus topology, all devices are connected to a single communication cable called the backbone cable. Data transmitted by one device travels along the shared cable. All connected devices can receive the transmitted signal, but only the intended device processes the data.

Basic structure



Bus topology requires termination at the ends of the backbone in traditional implementations to prevent signal reflection.

1.3 Cost Comparison

One of the major advantages of Bus topology is its relatively low implementation cost.

It requires:

- A single backbone cable
- Fewer cable connections
- Fewer networking devices
- Less installation material

Star topology requires an individual cable connection between every node and the central switch. Therefore, the amount of cabling and networking hardware is greater.

For example, if an organization has 20 computers, each computer in a Star topology requires a connection to the central switch. In contrast, Bus topology can connect multiple devices along one backbone.

Therefore:

Bus topology → Lower initial cost

Star topology → Higher initial cost

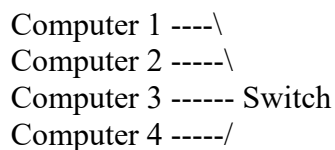
However, the initial cost is not the only factor. Star topology can be more economical in the long term because faults are easier to identify and network expansion is easier.

1.4 Reliability Comparison

Star topology provides better reliability than Bus topology in terms of individual node failures.

If one computer or its connecting cable fails in a Star network, the remaining computers can continue to communicate.

For example:



If Computer 3 fails, Computers 1, 2, and 4 can continue to communicate.

In Bus topology, all devices depend on the same backbone. If the backbone cable is damaged, communication across the network can be disrupted.

Therefore, Star topology provides better reliability.

1.5 Performance Comparison

Star topology generally provides better performance because each device has a dedicated connection to the switch.

Modern switches can provide:

- Dedicated communication paths
- Full-duplex communication
- Better traffic management
- Reduced collisions
- Higher bandwidth utilization

In a traditional shared Bus network, all devices use the same communication medium. As the

number of devices increases, network traffic also increases.

This can result in:

- Increased congestion
- More collisions
- Lower throughput
- Increased waiting time

Therefore, Star topology is generally superior in terms of performance.

1.6 Scalability Comparison

Star topology is easier to expand.

If a new computer needs to be added, it can simply be connected to an available switch port.

Existing:

```
PC1 ----\  
PC2 ----- Switch  
PC3 ----/
```

Expansion:

```
PC1 ----\  
PC2 ----- Switch ---- PC4  
PC3 ----/
```

In Bus topology, adding more devices increases the traffic on the shared backbone.

Therefore, Star topology provides better scalability.

1.7 Summary of Star vs. Bus

Feature	Star	Bus
Cost	Higher	Lower
Reliability	High	Low
Performance	High	Moderate/Low
Scalability	Good	Poor
Fault isolation	Easy	Difficult
Cable requirement	High	Low
Installation	Easy	Simple
Maintenance	Easy	Difficult
Central dependency	Yes	No central switch
Suitable for	Modern LANs	Small/simple networks

Conclusion

Star topology is generally preferable for modern networks because it provides better performance, reliability, scalability, and maintenance. Bus topology is advantageous mainly when low cost and simple installation are the primary requirements.

2. Mesh Topology vs. Star Topology

2.1 Mesh Topology

Mesh topology is a network structure in which devices are connected through multiple communication links.

There are two major types:

1. Full Mesh
2. Partial Mesh

In a full mesh topology, every node is directly connected to every other node.

For n nodes, the number of links required is:

$$\text{Links} = n(n - 1) / 2$$

For example, if there are 5 nodes:

$$5(5 - 1) / 2 = 10 \text{ links}$$

This demonstrates why full mesh networks can become expensive as the number of nodes increases.

2.2 Star Topology

In Star topology, each device communicates through a central switch.

Unlike Mesh topology, individual devices do not need direct connections to every other device.

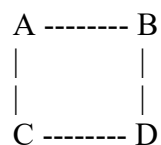
This makes Star topology considerably simpler.

2.3 Fault Tolerance

Fault tolerance refers to the ability of a network to continue operating even when one or more components fail.

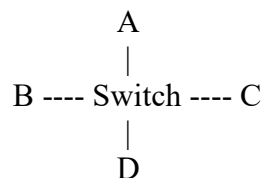
Mesh topology provides excellent fault tolerance.

Suppose two devices are connected through multiple paths:



If one link fails, another route may still be available.

Star topology has less redundancy.



If one cable fails, only that particular device is affected.

However, if the central switch fails, communication between all connected devices is interrupted.

Therefore:

Mesh → Very high fault tolerance

Star → Moderate fault tolerance

2.4 Reliability

Mesh topology is highly reliable because it has multiple communication paths.

It is particularly useful in environments where network downtime is unacceptable.

Examples include:

- Military networks
- Telecommunications

- Data centers
- Critical infrastructure
- Backbone networks

Star topology also provides good reliability, but its central switch creates a possible single point of failure.

2.5 Cost

Mesh topology is significantly more expensive.

The reasons include:

- Large number of cables
- More network interfaces
- More ports
- More configuration
- Greater maintenance requirements

Star topology requires fewer connections and therefore has lower implementation costs.

2.6 Complexity

Mesh topology is highly complex because multiple connections need to be established and managed.

Star topology is relatively simple.

Network administrators can easily identify which device is connected to which switch port.

Therefore:

Star → Simple

Mesh → Complex

2.7 Performance

Mesh topology can provide excellent performance because multiple paths are available.

Traffic can potentially be routed through alternative paths, reducing dependence on one communication link. Star topology also provides high performance, especially when modern switches are used. For most ordinary LAN environments, Star topology offers a better balance between performance, cost, and simplicity.

2.8 Summary of Mesh vs. Star

Feature	Mesh	Star
Reliability	Very High	High
Fault tolerance	Excellent	Moderate
Cost	Very High	Moderate
Complexity	Very High	Low
Scalability	Expensive	Good
Maintenance	Difficult	Easy
Redundancy	Excellent	Limited
Performance	Very High	High
Installation	Difficult	Easy
Suitable for	Critical networks	LANs and offices

Conclusion

Mesh topology is superior when maximum reliability and fault tolerance are required. Star topology is more practical for normal organizational networks because it provides an excellent balance between cost, performance, reliability, and simplicity.

3. Bus Topology vs. Mesh Topology

3.1 Installation Complexity

- Bus topology is one of the simplest topologies to install.
- It requires a shared backbone and connections to individual devices.
- Mesh topology requires multiple connections between nodes.
- As the number of devices increases, the number of links increases rapidly.
- Therefore, Bus topology is much easier to install.

3.2 Cost

Bus topology is less expensive because it requires:

- Less cable
- Fewer interfaces
- Fewer networking devices
- Less installation effort

Mesh topology is expensive because every node requires multiple links.

For example:

For 10 nodes in full mesh:

$$10 \times 9 / 2 = 45 \text{ links}$$

For 20 nodes:

$$20 \times 19 / 2 = 190 \text{ links}$$

This shows how rapidly the number of connections increases.

3.3 Reliability

Although Bus topology is inexpensive, it has lower reliability.

The entire network depends heavily on the backbone cable.

If the backbone fails:

Network communication can be disrupted for all devices.

Mesh topology provides multiple routes.

If one connection fails:

Traffic can potentially use another path.

Therefore, Mesh topology has significantly better reliability.

3.4 Scalability

Bus topology does not scale well.

Adding more devices causes:

- Increased traffic
- More contention
- Reduced performance
- Greater difficulty in troubleshooting

Mesh topology can accommodate additional nodes, but doing so becomes increasingly expensive and complex.

Therefore, both have limitations, but for different reasons.

3.5 Performance

Bus topology uses a shared medium.

As more devices are added, the available communication capacity is shared among more nodes.

Mesh topology can provide multiple communication paths and therefore can achieve high performance.

However, the exact performance depends on the routing mechanisms, links, hardware, and network design.

3.6 Maintenance

Bus topology is relatively easy to set up but can be difficult to troubleshoot. A fault in the backbone may affect multiple devices. Mesh topology has more individual connections, so identifying and maintaining links can become complicated. However, the redundancy of Mesh topology provides an important advantage: a single link failure does not necessarily cause network-wide failure.

3.7 Summary

Feature	Bus	Mesh
Cost	Very Low	Very High
Installation	Easy	Difficult
Reliability	Low	Very High
Fault tolerance	Poor	Excellent
Performance	Moderate/Low	High
Scalability	Poor	Limited by cost
Maintenance	Moderate/Difficult	Difficult
Redundancy	Very Low	Very High

Conclusion

Bus topology is suitable when simplicity and low cost are more important than reliability. Mesh topology is suitable for applications requiring high availability, redundancy, and fault tolerance.

4. Hybrid Topology vs. Star Topology

4.1 Hybrid Topology

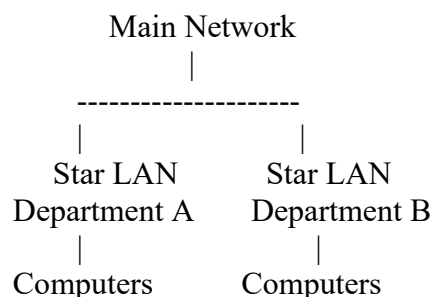
Hybrid topology is created by combining two or more different network topologies.

Examples include:

- Star-Bus
- Star-Ring
- Star-Mesh
- Mesh-Star

Hybrid topology is commonly used in large organizations because different departments may have different networking requirements.

For example:



4.2 Scalability

Hybrid topology provides excellent scalability.

Organizations can expand individual sections of the network without completely redesigning the entire infrastructure.

For example, one department may use Star topology while another department uses a different topology.

This flexibility makes Hybrid topology suitable for:

- Universities
- Hospitals
- Large companies
- Industrial organizations
- Government networks

4.3 Star Topology Scalability

Star topology is also scalable. Additional devices can be connected to the central switch if ports are available. However, if all ports are occupied, another switch must be added. Therefore, Star topology has good scalability, while Hybrid topology generally offers greater flexibility.

4.4 Flexibility

Hybrid topology has a major advantage in flexibility.

Different sections can use different topologies according to their requirements.

For example:

- Administration → Star
- Data center → Mesh
- Older section → Bus
- Backbone → High-speed redundant links

These sections can be integrated into a single larger network.

4.5 Cost

Hybrid topology is usually more expensive than a simple Star topology.

This is because it may require:

- Different networking devices
- More complex cabling
- Additional switches and routers
- Advanced configuration
- Skilled administrators

However, the higher cost may be justified by its scalability and flexibility.

4.6 Maintenance

Maintenance is more complex in Hybrid topology.

Administrators must understand different network structures.

A problem in one segment may require analysis of:

- Switches
- Routers
- Cables
- Network interfaces
- Different topology segments

Star topology is comparatively easier to troubleshoot.

4.7 Summary

Feature	Hybrid	Star
Flexibility	Excellent	Good
Scalability	Excellent	Good
Cost	High	Moderate
Complexity	High	Low
Maintenance	Moderate/High	Easy

Feature	Hybrid	Star
Reliability	High	High
Expansion	Very Flexible	Good
Configuration	Complex	Simple

Conclusion

Hybrid topology is more suitable for large and complex organizations where flexibility and expansion are important. Star topology is preferable for smaller and medium-sized networks where simplicity and easy maintenance are required.

5. Maintenance Comparison of Different Topologies

Maintenance is an important consideration when selecting a network topology. Maintenance includes:

- Identifying faults
- Replacing damaged cables
- Monitoring network traffic
- Managing network devices
- Troubleshooting connectivity problems
- Upgrading hardware
- Preventing network downtime

5.1 Star Topology Maintenance

Star topology is relatively easy to maintain.

Every device has an individual connection to the central switch.

If one computer stops communicating, the administrator can inspect:

1. The computer
2. Network interface
3. Ethernet cable
4. Switch port

Other computers can continue operating during troubleshooting.

Advantages

- Easy fault isolation
- Easy cable replacement
- Simple network monitoring
- Easy device addition
- Centralized management

Disadvantage

The central switch is critical. If it fails, many or all connected devices may lose network connectivity.

5.2 Bus Topology Maintenance

Bus topology can be difficult to troubleshoot. A single backbone cable is shared by multiple devices. If the backbone develops a fault, communication can be affected across the network.

The administrator may have to inspect a large portion of the cable to locate the problem.

Problems include:

- Difficult fault detection
- Backbone failure
- Signal problems

- Performance degradation
- Difficult expansion

Therefore, Bus topology has higher maintenance difficulty than Star topology.

5.3 Mesh Topology Maintenance

Mesh topology provides excellent reliability but creates significant maintenance complexity.

A large number of links must be monitored. If there are 10 devices in a full mesh, there are 45 links. If there are 20 devices, there are 190 links. Therefore, maintaining all these connections requires considerable effort. However, the major advantage is that failure of one link may not interrupt communication because alternate paths can exist.

5.4 Hybrid Topology Maintenance

Hybrid topology has moderate to high maintenance complexity.

Since multiple topologies are combined, administrators need to understand the structure of every network segment.

For example, if a university has:

- Star topology in laboratories
- Mesh connections in the server room
- Wireless networks in classrooms

the administrator needs to manage all these different components.

5.5 Maintenance Comparison

Topology	Maintenance Difficulty	Fault Isolation	Reliability
Star	Easy	Excellent	High
Bus	Difficult	Poor	Low
Mesh	Very Difficult	Moderate/Complex	Very High
Hybrid	Moderate/High	Depends on design	High

Conclusion

In conclusion, Star topology is easiest to maintain, while Bus topology is more difficult due to backbone failures. Mesh topology is highly reliable but complex to maintain, whereas Hybrid topology offers flexibility but requires skilled management.